

ELEC 221 Lecture 07

CT Fourier series and filters

Thursday 1 October 2026

Announcements

- Tutorial Assignment 2 on Monday: bring a short (3-5^{second}) music sample in .wav format
- Quiz 4 on Tuesday

Last time

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jkw t} \rightarrow y(t) = \sum_{k=-\infty}^{\infty} C_k \underbrace{H(jkw)}_{\text{transfer function (in our context, freq. response)}} e^{jkw t}$$

transfer function
(in our context,
freq. response)

Fourier synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jkw t} \quad \rightarrow \text{Fourier coeffs.}$$

Fourier analysis equation:

$$C_k = \frac{1}{T} \int_T e^{-jkw t} x(t) dt$$

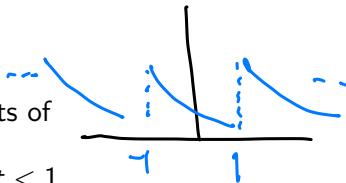
(integral over one period)

↓ periodic (+ additional cond.)

Last time

We evaluated the Fourier series coefficients of

$$x(t) = e^{-t}, \quad -1 \leq t < 1$$



We got an exact result

$$c_k = \frac{(-1)^k}{2(1 + jk\pi)} [e - e^{-1}]$$

Last time

Can we *always* express a signal as a Fourier series?

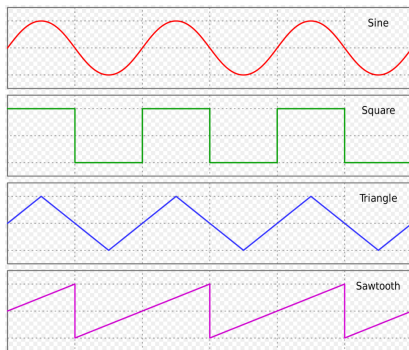


Image credit: *Sine, square, triangle, and sawtooth waveforms* (author: Omegatron)

https://en.wikipedia.org/wiki/Triangle_wave#/media/File:Waveforms.svg (CC BY-SA 3.0)

Last time

Given a **periodic function**, if over one period,

- ① is single-valued
- ② is absolutely integrable
- ③ has a finite number of maxima and minima
- ④ has a finite number of discontinuities

} *bounded variation*

then the Fourier series converges to

- $x(t)$ where it is continuous
- half the value of the jump where it is discontinuous

Today

Learning outcomes:

- Evaluate the Fourier series coefficient of a square wave; describe the Gibbs phenomenon
- State and apply key properties of CT Fourier series
- Express filters using the frequency response

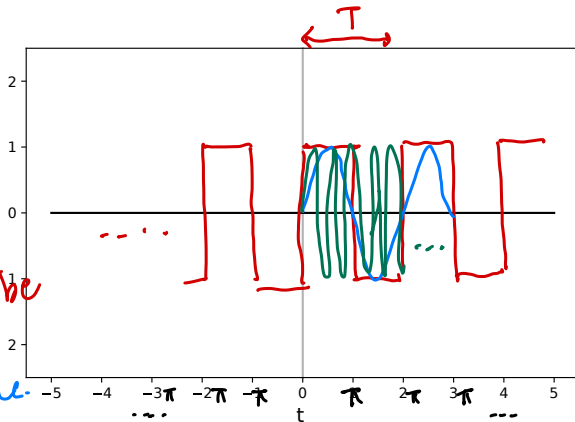
Example: the square wave

$$\sum_k c_k e^{jk\omega t}$$
$$a_0 + \sum_k a_k \cos(k\omega t) + \sum_k b_k \sin(k\omega t)$$
$$x(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & \pi < t < 2\pi \end{cases}$$

$$T = 2\pi \Rightarrow \omega = 1$$

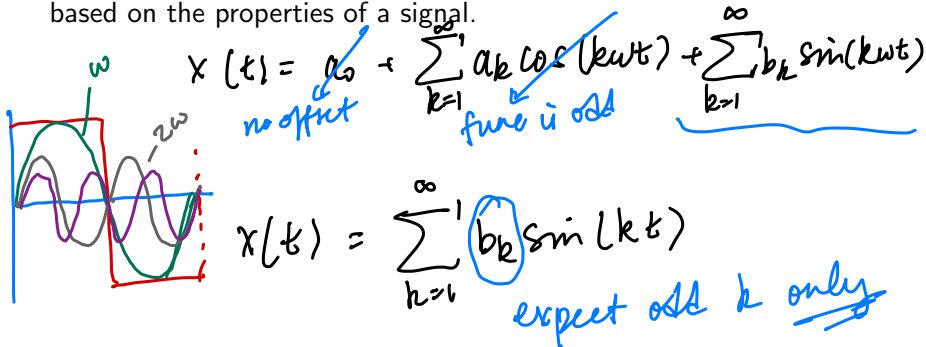
find c_k

or
 ~~a_0, a_k~~ b_k
these will be there
odd func.



Example: the square wave

Evaluate its Fourier coefficients. We can often take shortcuts based on the properties of a signal.



You can derive that for a 2π -periodic function, the coefficients have the following form:

$$\frac{1}{T} \int_T e^{j(k-l)\omega t} dt = \begin{cases} 1 & \text{if } k=l \\ 0 & \text{otherwise} \end{cases}$$

k, l ints. (pointing to k and l)

$$b_k = \frac{1}{\pi} \int_0^{2\pi} x(t) \sin(kt) dt$$

$$\begin{aligned}
 x(t) &= \sum_{k=-\infty}^{\infty} b_k \sin(kt) \\
 \int_0^{2\pi} \sin(mt) x(t) dt &= \int_0^{2\pi} \sum_{k=-\infty}^{\infty} b_k \sin(kt) \sin(mt) dt \\
 &= \sum_{k=-\infty}^{\infty} b_k \int_0^{2\pi} \sin(kt) \sin(mt) dt
 \end{aligned}$$

$k=m$ gives

$$\begin{aligned}
 a_k &= 2 \cdot \operatorname{Re}(C_k) \\
 &= 2 \cdot \frac{1}{2\pi} \int_0^{2\pi} \text{stuff} \cdot dt \\
 &= \frac{1}{\pi} \int_0^{\pi} \text{stuff} \cdot dt
 \end{aligned}$$

Example: the square wave

Since the function is symmetric, we can rewrite this:

$$b_k = \frac{1}{\pi} \int_0^{2\pi} x(t) \sin(kt) dt$$

$$= \frac{2}{\pi} \int_0^{\pi} x(t) \sin(kt) dt$$

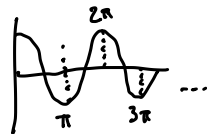
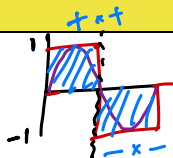
$$= \frac{2}{\pi} \int_0^{\pi} \sin(kt) dt$$

$$= \frac{2}{\pi} \left(-\frac{1}{k} \right) \cos(kt) \Big|_0^{\pi}$$

$$= -\frac{2}{\pi k} [\cos(k\pi) - 1]$$

$$= -\frac{2}{\pi k} [(-1)^k - 1]$$

$$= \begin{cases} 0 & \text{if } k \text{ even} \\ \frac{4}{\pi k} & \text{if } k \text{ odd} \end{cases}$$



$$\cos(k\pi) = \begin{cases} -1 & \text{if } k \text{ odd} \\ 1 & \text{if } k \text{ even} \end{cases}$$

$$= (-1)^k$$

Will it converge?

Convergence: signal power and energy

Define the **total energy** of a signal over the period t_1 to t_2 as

$$E = \int_{t_1}^{t_2} |x(t)|^2 dt$$

The time-averaged **power** is

$$P = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

relate to C_k

Signals with finite energy over a period can be represented using a Fourier series.

Convergence: signal power and energy

$$x(t) = \sum_k c_k e^{jk\omega t}$$

First, derive an important relationship between our signal and its

Fourier coefficients (**Parseval's relation**): $|x(t)|^2 = x(t) \cdot x^*(t)$

$$\begin{aligned} P_T &= \frac{1}{T} \int_T |x(t)|^2 dt \\ &= \frac{1}{T} \int_T \left(\sum_{k=-\infty}^{\infty} c_k e^{jk\omega t} \right) \left(\sum_{l=-\infty}^{\infty} c_l^* e^{-jl\omega t} \right) dt \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} c_k c_l^* \int_T e^{j(k-l)\omega t} dt \\ &= \sum_{k=-\infty}^{\infty} c_k c_k^* \end{aligned}$$

*k=l survives.
= ~~T~~*

$$\boxed{\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |c_k|^2}$$

Parseval's relation.

Convergence: signal power and energy

Imagine truncating the sum at N :

$$x_N(t) = \sum_{k=-N}^N c_k e^{jk\omega t}$$

Consider approximation error of the truncation:

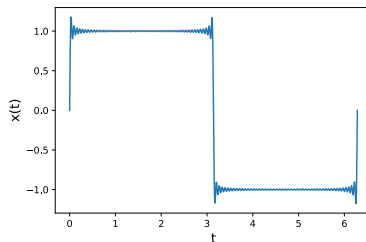
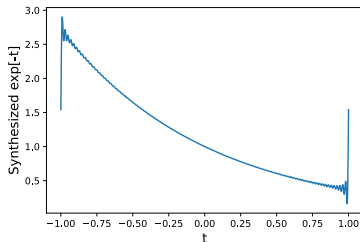
$$e_N(t) = x(t) - x_N(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t} - \sum_{k=-N}^N c_k e^{jk\omega t}$$

How much energy in error is there over a given period? $= \sum_{|k|>N} c_k e^{jk\omega t}$

$$E_N = \int_T |e_N(t)|^2 dt = T \cdot \underbrace{\sum_{|k|>N} |c_k|^2}$$

Take limit as $N \rightarrow \infty \dots$ $E_N \rightarrow 0$

Gibb's phenomenon



The amount of ringing, or “overshoot”, is about 9% of the jump of the discontinuity, no matter where we truncate.

Watch a derivation here:

<https://www.youtube.com/watch?v=MvXY8MXSit4>

Properties of Fourier series

In the square wave example, we leveraged some shortcuts to compute the Fourier coefficients.

They have other useful properties that help evaluate them.

Let's see what happens to the Fourier series when we apply:

- Superposition
- Time shift / scale / reversal
- Multiplication

Properties of Fourier series

Fourier coefficients combine linearly.

Suppose we have two signals $x(t), y(t)$ with period T ,

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t},$$

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega t},$$



Then the signal $z(t) = Ax(t) + By(t)$ has the form

$$z(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$c_k = Aa_k + Bb_k$$

Properties of Fourier series

Time shift $x(t) \rightarrow x(t - t_0)$:

$$\begin{aligned}
 c_k' &= \frac{1}{T} \int_T x(t - t_0) e^{-jk\omega t} dt \quad \text{change} \\
 &= \frac{1}{T} \int_T x(\tau) e^{-jk\omega(\tau + t_0)} d\tau \quad \tau = t - t_0 \\
 &= e^{-jk\omega t_0} \cdot \underbrace{\frac{1}{T} \int_T x(\tau) e^{-jk\omega \tau} d\tau}_{C_k} \\
 &= e^{-jk\omega t_0} \cdot C_k
 \end{aligned}$$

Properties of Fourier series

Time scale $x(t) \rightarrow x(\alpha t)$.

If original period was T , new period is T/α :

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$x(\alpha t) = \sum_{k=-\infty}^{\infty} c_k e^{j k \omega \alpha t}$$

Properties of Fourier series

Multiplication leads to convolution of the coefficients:

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t},$$

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega t},$$

Then the signal $z(t) = x(t)y(t)$ has the form

$$z(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$c_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l}$$

convolution of a_k, b_k !

Properties of Fourier series

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left. \begin{array}{l} x(t) \\ y(t) \end{array} \right\} \begin{array}{l} \text{Periodic with period } T \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/T \end{array}$	$\begin{array}{l} a_k \\ b_k \end{array}$
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} = e^{jM(2\pi/T)t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a_{-k}^*
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t - \tau)d\tau$	$T a_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk \frac{2\pi}{T} a_k$
Integration		$\int_{-\infty}^t x(t) dt$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right) a_k = \left(\frac{1}{jk(2\pi/T)}\right) a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \mathcal{E}\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\begin{cases} \Re\{a_k\} \\ j\Im\{a_k\} \end{cases}$

Exercise

Go back to the square wave

Homework.

$$x(t) = \begin{cases} 1, & 0 < t < \pi, \\ -1, & \pi < t < 2\pi \end{cases}$$

We obtained

$$x(t) = \sum_{k=1}^{\infty} b_k \sin(kt), \quad b_k = \begin{cases} 0, & k \text{ is even} \\ 4/k\pi, & k \text{ is odd} \end{cases}$$

What are the Fourier coefficients of the square wave

$$x(t) = \begin{cases} 1, & -\frac{\pi}{4} < t < \frac{\pi}{4}, \\ -1, & \frac{\pi}{4} < t < \frac{3\pi}{4} \end{cases}$$

Exercise

Step 1: express the b_k as the “original” coefficients c_k

Exercise

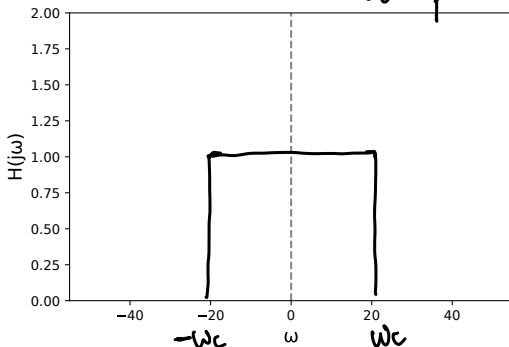
Step 2: apply the transformations

Example

What does a system with the following frequency response do?

$$H(j\omega) = \begin{cases} 1 & |\omega| \leq \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

low pass filter



Filters

Filters are LTI systems that can be used to separate out, combine, or modify the components of a signal at specific frequencies.

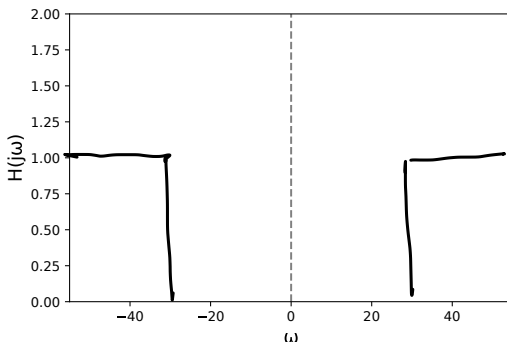
Two key types:

- **Frequency-shaping**: change the amplitudes of parts of a signal at specified frequencies
- **Frequency-selective**: eliminate or attenuate parts of a signal at specified frequencies

CT frequency-selective filters

We can also consider an ideal **highpass filter**:

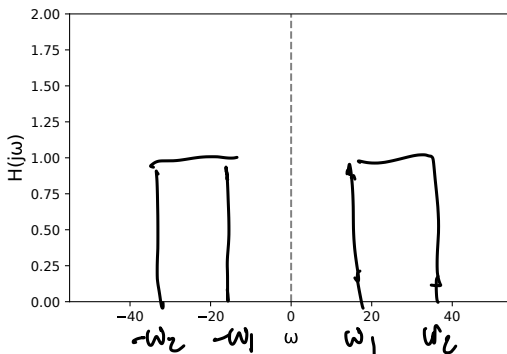
$$H(j\omega) = \begin{cases} 1 & |\omega| \geq \omega_c \\ 0 & |\omega| \leq \omega_c \end{cases}$$



CT frequency-selective filters

Or an ideal **bandpass** filter:

$$H(j\omega) = \begin{cases} 1 & \omega_1 \leq |\omega| \leq \omega_2 \\ 0 & \text{otherwise} \end{cases}$$



For next time

Content:

- DT periodic signals
- DT Fourier series coefficients
- DT frequency response and filters

Action items:

- ① Work on practice assignments 1 and 2
- ② Choose music for tutorial assignment

Recommended reading:

- From today's class: Oppenheim 3.3-3.5, 3.9
- Suggested problems: 3.1, 3.3-3.8, 3.13, 3.15, 3.19, 3.21, 3.22a,c, 3.23-3.26, 3.33, 3.35, 3.43-3.46
- For next class: Oppenheim 3.6-3.7